

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application

Practical – X

Roll No.: T013	Name: Saqlain Khan
Class: TYIT	Batch: 1
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A) Create a PWA with a Web App Manifest file including:

- App name
- Icons
- Start URL
- Display mode
- Implement Push Notifications in a PWA.

CODE :-

manifest.json :-

```
{  
  "name": "My PWA Application",  
  "short_name": "MyPWA",  
  "start_url": "/",  
  "display": "standalone",  
  "background_color": "#ffffff",  
  "theme_color": "#0d6efd",  
  "icons": [  
    {  
      "src": "icons/icon1.png",  
      "sizes": "192x192",  
      "type": "image/png"  
    },  
    {  
      "src": "icons/icon2.png",  
      "sizes": "512x512",
```

```
    "type": "image/png"
  }
]
}
```

index.html :-

```
<!DOCTYPE html>
<html>
<head>
  <title>My PWA</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color" content="#0d6efd">
</head>
<body>
  <h1>My Progressive Web App</h1>
  <button onclick="showNotification()">
    Show Notification
  </button>
  <script src="app.js"></script>
</body>
</html>
```

app.js :-

```
if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('service-worker.js')
    .then(() => console.log('Service Worker Registered'));
}

function showNotification() {
  Notification.requestPermission().then(permission => {
    if (permission === 'granted') {
      navigator.serviceWorker.ready.then(registration => {
```

```
        registration.showNotification('PWA Notification', {
            body: 'Push Notification Working Successfully!',
            icon: 'icons/icon1.png'
        });
    });
}
});
}
```

service-worker.js :-

```
if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('service-worker.js')
        .then(() => console.log('Service Worker Registered'));
}

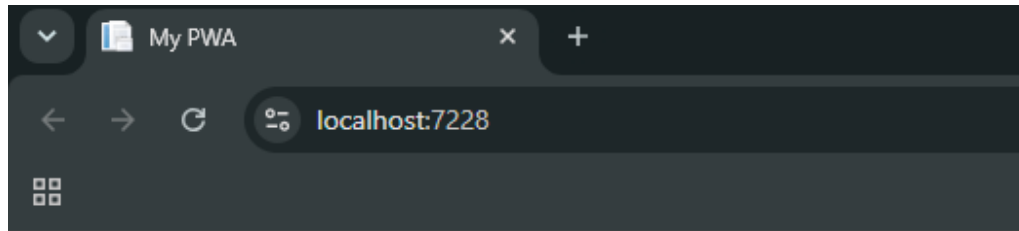
function showNotification() {
    Notification.requestPermission().then(permission => {
        if (permission === 'granted') {
            navigator.serviceWorker.ready.then(registration => {
                registration.showNotification('PWA Notification', {
                    body: 'Push Notification Working Successfully!',
                    icon: 'icons/icon1.png'
                });
            });
        }
    });
}
```

Program.cs :-

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
```

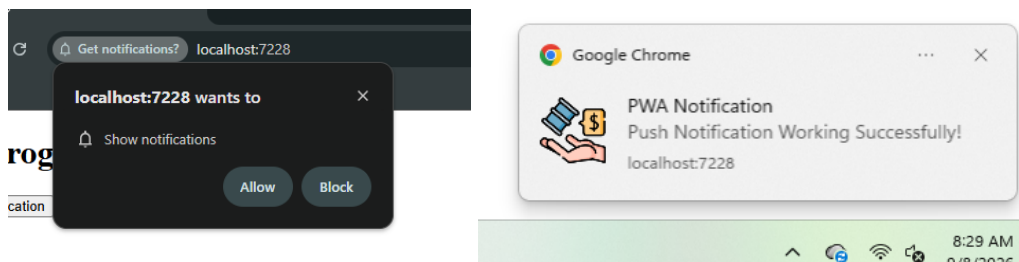
```
app.Run();
```

OUTPUT :-



My Progressive Web App

Show Notification



B) Develop an E-Commerce PWA with Products Listing and Offline Cart Personality

CODE :-

Program.cs :-

```
var builder = WebApplication.CreateBuilder(args);  
var app = builder.Build();  
app.UseDefaultFiles();  
app.UseStaticFiles();  
app.Run();
```

manifest.json :-

```
{  
  "name": "E-Commerce PWA",  
  "short_name": "ShopPWA",  
  "start_url": "/index.html",  
  "display": "standalone",
```

```
"background_color": "#ffffff",
"theme_color": "#0d6efd",
"icons": [
  {
    "src": "icons/icon1.png",
    "sizes": "192x192",
    "type": "image/png"
  },
  {
    "src": "icons/icon2.png",
    "sizes": "512x512",
    "type": "image/png"
  }
]
```

index.html :-

```
<!DOCTYPE html>
<html>
<head>
  <title>E-Commerce PWA</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color" content="#0d6efd">
  <style>
    body {
      font-family: Arial;
      margin: 20px;
    }

    .container {
      display: flex;
```

```
        gap: 30px;
    }

    .cart-section {
        width: 30%;
        border: 1px solid gray;
        padding: 15px;
        min-height: 300px;
    }

    .product-section {
        width: 70%;
    }

    .products {
        display: flex;
        flex-wrap: wrap;
        gap: 15px;
    }

    .product {
        border: 1px solid gray;
        padding: 10px;
        width: 200px;
    }

    button {
        padding: 5px 10px;
        cursor: pointer;
    }
</style>
</head>
<body>
    <h1>E-Commerce PWA</h1>
    <div class="container">
```

```
<!-- Cart Section -->

<div class="cart-section">

  <h2>Cart Items</h2>

  <ul id="cartList"></ul>

</div>

<!-- Product Section -->

<div class="product-section">

  <h2>Products</h2>

  <div class="products">

    <div class="product">

      <h3>Laptop</h3>

      <p>Price: ₹50000</p>

      <button onclick="addToCart('Laptop')">

        Add to Cart

      </button>

    </div>

    <div class="product">

      <h3>Mobile</h3>

      <p>Price: ₹20000</p>

      <button onclick="addToCart('Mobile')">

        Add to Cart

      </button>

    </div>

    <div class="product">

      <h3>Headphones</h3>

      <p>Price: ₹3000</p>

      <button onclick="addToCart('Headphones')">

        Add to Cart

      </button>

    </div>

  </div>

</div>
```

```
        </div>
    </div>
</div>
<script src="app.js"></script>
</body>
</html>
```

app.js :-

```
if ('serviceWorker' in navigator)
{
    navigator.serviceWorker.register('service-worker.js')
        .then(() => console.log('Service Worker Registered'));
}
let cart = JSON.parse(localStorage.getItem('cart')) || [];
displayCart();
function addToCart(product)
{
    cart.push(product);
    localStorage.setItem('cart', JSON.stringify(cart));

    displayCart();
}
function displayCart()
{
    let cartList = document.getElementById('cartList');

    cartList.innerHTML = "";
    cart.forEach(item =>
    {
        let li = document.createElement('li');
        li.textContent = item;
```

```
        cartList.appendChild(li);
    });
}

service-worker.js :-
const CACHE_NAME = 'ecommerce-cache-v1';
const urlsToCache = [
    '/',
    '/index.html',
    '/app.js',
    '/manifest.json'
];

self.addEventListener('install', event =>
{
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(cache =>
                {
                    return cache.addAll(urlsToCache);
                }
            )
    );
});

self.addEventListener('fetch', event =>
{
    event.respondWith(
        caches.match(event.request)
            .then(response =>
                {
                    return response || fetch(event.request);
                }
            )
    );
});
```

});

OUTPUT :-

E-Commerce PWA

Cart Items

- Laptop
- Mobile
- Headphones

Products

Laptop

Price: ₹50000

Add to Cart

Mobile

Price: ₹20000

Add to Cart

Headphones

Price: ₹3000

Add to Cart

E-Commerce PWA

Cart Items

- Laptop
- Mobile
- Headphones

Products

Laptop

Price: ₹50000

Add to Cart

Mobile

Price: ₹20000

Add to Cart

Headphones

Price: ₹3000

Add to Cart

